

Part-A:Task 4.1-Transformers & Attention Explained

What Problems Did Transformers Solve?

Earlier language models processed text one word at a time. This made it difficult for them to remember information from much earlier in a sentence or paragraph, so they often missed important relationships between words. **Transformers** solved this problem by looking at all the words in a sentence together instead of reading them one by one. This helps the model understand the overall meaning and connect words even when they are far apart.

What Is Attention?

Attention is the main idea behind transformers. It helps the model decide which words are most important when understanding another word. Instead of giving equal importance to every word, the model focuses more on the words that provide useful context.

For example, in the sentence "**The animal didn't cross the street because it was too tired,**" attention helps the model understand that "**it**" refers to "**the animal**" because being tired makes sense for an animal, not for a street.

What Is the Context Window?

A transformer can only process a limited amount of text at one time. This limit is called the **context window**. If the input is longer than the context window, some information may be left out. Longer inputs also require more computation, which increases the processing cost. That is why keeping prompts clear and concise is important when working with Large Language Models (LLMs).